



KENYATTA UNIVERSITY

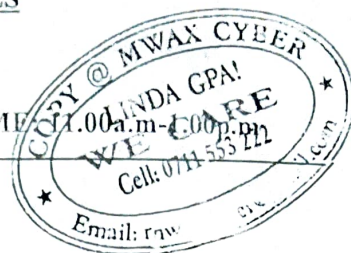
UNIVERSITY EXAMINATIONS 2017/2018

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (MECHANICAL ENGINEERING)

EMM 107: ENGINEERING MATERIALS

DATE: Tuesday 28th August 2018

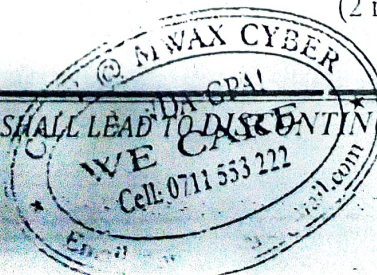
TIME



INSTRUCTIONS:

Answer any *THREE* questions

- ✓ 1. a) What are the reasons for adding alloying elements to steel? (2 marks)
- b) Describe the two means of forming polymers. (10 marks)
- c) State the main alloying elements in:
- i) Brass
 - ii) Bronze
 - iii) Duralumin (3 marks)
- d) What alloying elements tend to form stable carbides within steel? (2 marks)
- e) What alloying elements tend to make carbides unstable within steel? How do they make carbides unstable? (3 marks)
2. a) Explain, with the aid of diagrams, the effects of the following elements to a plain carbon steel
- i) Nickel
 - ii) Manganese (15 marks)
- b) What are some of the attractive engineering properties of aluminium and aluminium alloys? (5 marks)
3. a) What are the attractive engineering properties of copper-nickel alloys that make them good materials for heat exchangers? (4 marks)
- b) What are the principal difficulties encountered during the manufacture of magnesium-based alloys? (2 marks)



c) What factors has made titanium and its alloys important in aerospace applications? (4 marks)

d) What are some reasons why additive agents are incorporated into plastics? (4 marks)

e) What are some major limitations to the extensive use of composite materials in engineering applications? (6 marks)

4. a) What are some attractive and limiting features of structural ceramics? (3 marks)

b) Why are enamels and glazes added to the surface of ceramics? (3 marks)

c) You want to produce a component from an elastomer. Should you vulcanise the rubber before or after the forming operation? (2 marks)

d) Why are few forming techniques used for thermosetting polymers? (3 marks)

e) Why are fillers added to polymers? (4 marks)

f) How are thermosetting polymers formed? (5 marks)

✓ 5. a) Describe the mechanism of corrosion of a metal by a liquid. (8 marks)

b) List three methods of Corrosion control (3 marks)

c) What are the drawbacks of using high purity metals to control corrosion? (2 marks)

d) Relate the changes in microstructure to the properties of steels as the carbon content increases from 0.1 to 1.2 percent. (5 marks)

e) Where do maraging steels find applications? (2 marks)

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SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN MECHANICAL ENGINEERING AND BACHELOR OF SCIENCE
(AEROSPACE ENGINEERING)

EMM 107/ EAR 103: ENGINEERING MATERIALS

DATE: WEDNESDAY 7TH JUNE 2017

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS:

ANSWER ANY THREE QUESTIONS

1. a) Distinguish between malleable cast iron and nodular iron in terms of their production and characteristics. (6 marks)
- b) Using the case of two elements A and B that are mutually soluble in the liquid and solid states, illustrate the manner in which a thermal-equilibrium diagram is constructed. (6 marks)
- c) Explain the effect of silicon on the properties of cast iron. (5 marks)
- d) What is the basic constitution of bronze? Distinguish between plain tin bronzes and phosphor bronze. (3 marks)
2. a) Describe the two ways of forming polymers. (10 marks)
- ✓ b) What are the major determinants of the properties of composite materials? (5 marks)
- ✓ c) What are the properties of aluminium and aluminium alloys that make them attractive for engineering applications? (5 marks)
2. a) Relate the changes in microstructure to the properties of steels as the carbon content increases from 0.1 to 1.2 percent. ✓ (10 marks)
- b) Give an outline of the factors that influence the choice of nickel-based alloys for high temperature engineering. (4 marks)
- c) List four methods of corrosion control. (4 marks)
- d) What are the drawbacks of using high purity metals to control corrosion? (2 marks)

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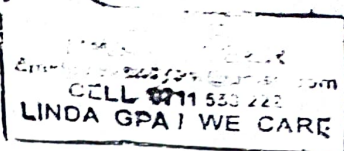
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4. a) What are the features that influence the properties of the fibre-reinforced composites. (5 marks)
- b) What properties make superalloys find use in jet engine, rocket and nuclear applications? (5 marks)
- c) What alloying elements tend to form stable carbides within steel? (3 marks)
- d) Which properties of titanium and its alloys are important in considering them for aerospace applications? (3 marks)
- e) What are the principal objectives of adding alloying elements to steel? (2 marks)
- f) List two reasons that make additive agents be incorporated into plastics (2 marks)
- g) Describe the effects of alloying elements on the polymorphic transformation temperatures. (9 marks)

What are the limitations in the use of composite materials in engineering? (6 marks)

A steel to be used as a spring is estimated to contain 10% Fe_3C and 90% ferrite. Can we consider this steel to be an eutectoid steel? (3 marks)

An as-received bar stock is observed to contain about 95% pearlite and 5% primary Fe_3C . Calculate the carbon content of the steel. (2 marks)



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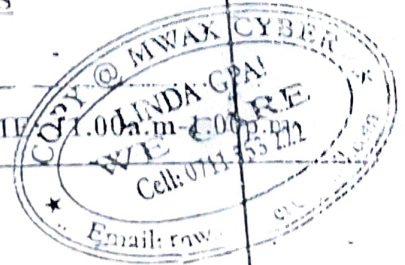
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